

DANG TRONG PHUC

Java Backend Engineer · Spring Boot · Microservices · Distributed Systems
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PROFESSIONAL SUMMARY

Java Backend Engineer with 2+ years of professional experience and additional hands-on project work building scalable, high-availability microservices and distributed systems. Core expertise in Java/Spring Boot (Spring MVC, Spring Security, Spring Data JPA, Spring Cloud), RESTful API design, Kafka/RabbitMQ event-driven architecture, and Redis caching. Solid foundation in OOP, SOLID principles, and Design Patterns; practises TDD with JUnit 5 and Mockito. Comfortable in Agile/Scrum environments (Jira, GitLab CI); familiar with Docker-based deployments on Linux. Quick learner with a can-do attitude; reads and communicates effectively in English technical environments (B1 Aptis, improving to B2).

TECHNICAL SKILLS

Languages: Java 8, 11, 17, 21 (primary); JavaScript / TypeScript (basic); SQL

Spring Ecosystem: Spring Boot 3, Spring MVC, Spring Data JPA/Hibernate, Spring Security + JWT, Spring Cloud (Gateway, Eureka, Config Server), Spring AOP

Architecture: Microservices, Event-Driven Architecture, API Gateway, Service Discovery, OOP, SOLID, Design Patterns (Repository, Factory, Builder, Strategy, Observer, Anti-Corruption Layer)

Messaging: Apache Kafka (topics, consumer groups, exactly-once semantics), RabbitMQ (JMS, pub/sub, retry with backoff)

Databases & Cache: MySQL, PostgreSQL; Redis (cache-aside, session, pub/sub, rate-limiting, distributed locking)

API & Docs: RESTful API (Spring MVC), gRPC / Protobuf (internal services), OpenAPI 3 / Swagger UI

Testing: JUnit 5, Mockito (TDD / unit testing), Postman (integration testing), load testing (10k+ concurrent)

DevOps & Tools: Docker, Docker Compose, GitLab CI/CD, Linux (Ubuntu/CentOS), Git, Maven, Jira, Postman, SonarQube

Resilience: Resilience4j Circuit Breaker, Token Bucket rate limiting (Bucket4j), idempotency keys, optimistic/pessimistic locking

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

Sep 2018 – Dec 2023

Bachelor of Engineering · Computer Electronic Engineering · Ho Chi Minh City

WORK EXPERIENCE

GIHOT STUDIO · Java Backend Developer

04/2024 – Present · Ho Chi Minh City

Building backend services for real-time multiplayer mobile games in a 20+ engineer Agile team. Focus on Java/Spring Boot microservices, internal tooling APIs, performance optimization, and production reliability.

- Designed and built **Spring Boot microservices** for GM Tool (admin dashboard), game configuration management, and real-time operational monitoring; all REST APIs documented with OpenAPI 3 / Swagger UI.
- Implemented Spring Security + JWT with RBAC; applied row-level ownership checks and API endpoint hardening across all internal services.
- Applied OOP, SOLID principles, and Design Patterns (Repository, Service Layer, Strategy) throughout Java codebase; wrote JUnit 5 + Mockito unit tests for core business logic; enforced code quality via SonarQube.
- Optimized Redis caching (cache-aside, TTL tuning) and RabbitMQ async pipelines — reduced DB load by ~30% and API response time by ~20%.
- Profiled and tuned JPA/Hibernate queries (N+1 resolution, composite indexing, batch fetching); conducted load tests at 10k+ virtual users; achieved 25% latency reduction through systematic bottleneck analysis.
- Participated in GitLab CI Agile sprints with Jira; collaborated with QA, DevOps, and client teams; resolved critical production incidents with structured root cause analysis.

Stack: Java 8/21, Spring Boot 3, Spring MVC, Spring Security, JWT, Spring Data JPA/Hibernate, RabbitMQ, Redis, MySQL, Docker, GitLab CI, Jira, Maven, SonarQube

HAHALOLO TRAVEL SOCIAL NETWORK · Java Backend Developer

06/2022 – 05/2023 · Ho Chi Minh City

Part-time backend developer during final university years on a high-traffic social travel platform (~10+ engineers, Agile/Scrum). Built RESTful APIs, implemented security, and optimized database performance.

- Designed and built **scalable RESTful APIs** using Java 8, Spring Boot, Spring MVC, and Spring Data JPA for core platform features (user profiles, trip posts, social feeds); documented all APIs with Swagger/OpenAPI 3.

- Implemented Spring Security + JWT with RBAC; secured all platform endpoints and managed tiered user access control.
- Applied OOP, SOLID, and Repository/Service Layer design patterns; wrote JUnit + Mockito unit tests for critical business logic; participated in peer code reviews.
- Profiled complex SQL queries and JPA/Hibernate data access layers (N+1 fix, index tuning, batch fetching) — reduced average query latency by ~40%.
- Worked alongside QA and DevOps teams in Docker/Linux environments for deployment, production log diagnosis, and rapid bug resolution.

Stack: Java 8, Spring Boot, Spring MVC, Spring Security, JWT, JPA/Hibernate, MySQL, PostgreSQL, Docker, Linux, Git, Maven, Swagger/OpenAPI

Self-directed period (06/2023 – 04/2024): Completed final-year thesis and graduated PTIT (Dec 2023); used remaining time for personal backend projects and deepening Java/distributed systems knowledge before joining Gihot Studio.

KEY PROJECTS

FPM — Financial Portfolio Manager (Personal, Cloud-Native Microservices)

10/2025 – Present

- Architected a **10-service cloud-native system** (User-Auth, Wallet, Transaction Engine, Reporting, Notification, OCR, AI, API Gateway, Eureka, Config Server) using Java 21, Spring Boot 3.5, Spring Cloud 2025.0; enforced strict domain boundaries and DDD principles.
- Designed dual-broker event pipeline: Kafka (8 topics, exactly-once transaction streaming) + RabbitMQ (domain event routing: wallet.created, balance.changed, budget.alerts) — fully decoupled async architecture.
- Built **9 shared internal libraries** (fpm-security, fpm-grpc, fpm-proto, fpm-messaging, fpm-testing, etc.) to standardize JWT, exception handling, response format, and event publishing across all services.
- Implemented gRPC inter-service layer with 6 Protobuf contracts + Anti-Corruption Layer; sub-50ms internal RPC via HTTP/2 + binary Protobuf.
- Applied Resilience4j Circuit Breaker (50% threshold), Token Bucket rate limiting, RabbitMQ retry with backoff, and Redis multi-layer caching (TTL 5 min) — ~35% DB read load reduction.
- Implemented dual-layer JWT security: Gateway validates token + Redis blacklist; downstream re-validates via shared fpm-security library (defense-in-depth vs. Mass Assignment); prevented race conditions using optimistic/pessimistic locking on wallet balance updates.
- Documented all APIs with OpenAPI 3 / Swagger UI; applied TDD with JUnit 5 + Mockito; configured Docker Compose + GitLab CI pipeline with SonarQube quality gates.

Stack: Java 21, Spring Boot 3.5, Spring Cloud, Kafka, RabbitMQ, Redis 7, PostgreSQL, gRPC/Protobuf, Docker Compose, GitLab CI, JUnit 5, Mockito, Swagger/OpenAPI 3

Sieu Anh Hung Dai Chien — Real-Time Card Battle Game (Java Backend)

04/2024 – 11/2024

- Built Spring Boot microservices backend for real-time multiplayer card game; applied SOLID and Design Patterns throughout; reduced system bottlenecks by 25% under 5,000+ concurrent users in load tests.
- Implemented reward distribution, mission tracking, and session management engines with strict transactional state consistency; secured endpoints with Spring Security + JWT.
- Developed GM Tool REST APIs with OpenAPI 3 / Swagger docs; applied JUnit + Mockito test coverage on core business logic.

Stack: Java 8, Spring Boot, Spring Security, JWT, Protobuf, Redis, MySQL, JUnit, Mockito, Swagger/OpenAPI

ADDITIONAL INFORMATION

English: B1 Aptis Certificate (British Council) — actively improving to B2; reads technical documentation and communicates effectively in English-speaking global teams.

Methodology: Agile / Scrum · GitLab + Jira · sprint planning, daily standup, retrospective, code review.

Focus: Clean architecture, TDD, distributed systems design, SQL/JPA performance profiling, proactive self-learner.